→ Shows how: 1) limits replaced infinitesimals 2) $\varepsilon-\delta$ definitions

4. Journey Through Genius — William Dunham

Dialogue Three: Convergent Sequence

Author: Let us prove the following

Theorem: If a sequence has a limit, it is bounded.

Sen: **Proof:** (My 1st attempt) Let, (y_n) be a given sequence having a limit a. From Defn(1) of Dialogue(2), for any arbitrary positive ε; there exists a natural number N such that for all $n > N$, $|y_n - a| < \varepsilon$.

$\therefore$ Either, $y_n - a < \varepsilon$ or $-(y_n - a) < \varepsilon$. $\therefore$ $y_n - \varepsilon < a < y_n + \varepsilon$

Or, $\boxed{a-\varepsilon < y_n < a+\varepsilon}$ → This relation holds for any arbitrary positive ε.

- $\forall \epsilon > 0, \exists N \in$ Nat no, s.t. $\forall n > N$, $\boxed{a-\epsilon < y_n < a+\epsilon}$

Q) Does there exist a value of ϵ where $N = 0$?

Author: We assume that a is the limit of a sequence (y_n). Now take an arbitrary value of ε greater than 0. According to the definition of the limit, the selected ε can always be related to N such that for all $n > N$, $|y_n - a| < \varepsilon$. Hence, starting with $n = N+1$, all the subsequent terms of the sequence satisfy the following inequalities $\boxed{a-\varepsilon < y_n < a+\varepsilon}$